

Structural Monitoring of Heritage Walls: Insights from the Medieval Walls of Gubbio Using Time-Series Decomposition

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Abstract. Historical masonry structures face increasing threats from changing weather patterns and seismic activity, particularly heritage walls suffering from insufficient maintenance. This study investigates the structural behavior of Gubbio's medieval walls in Italy, analyzing correlations between inclination variations and seismic events using a 3-year dataset from Structural Health Monitoring stations. The research applies a hybrid time-series decomposition methodology to isolate trends, seasonality, and residuals of the measurements, while integrating CUSUM analysis to detect potential structural shifts. Seismic data from the National Institute of Geophysics and Volcanology is processed and overlaid with the residuals in order to identify possible correlations. Results reveal that the medieval walls exhibit complex multi-timescale response mechanisms, with residuals showing significant excursions beyond statistical control limits and a persistent directional bias. Notably, substantial cumulative deviations occur in temporal proximity to seismic events, suggesting these structures manifest delayed rather than immediate responses to seismic excitations. The study demonstrates the utility of advanced time-series analysis techniques for enhancing heritage structure monitoring, providing a replicable Python-based workflow framework. This research contributes to heritage conservation by enabling detection of subtle structural changes, while establishing a foundation for future studies incorporating finite element modeling and multivariate analysis to better understand complex environmental-structural interactions in historically significant masonry structures.

Keywords: Structural health monitoring, Medieval masonry walls, Time-series decomposition, Heritage preservation, Seismic response analysis

1 Introduction

Time series data is spread through numerous aspects of daily life, from financial markets and economic indicators to environmental measurements and biological signals. These sequential observations, indexed in time order, serve as foundational

representations of dynamic real-world phenomena across scientific disciplines and industrial applications [1]. The mathematical and computational frameworks developed for time series analysis have evolved significantly to address the inherent complexities within temporal data, enabling researchers and practitioners to extract meaningful insights from seemingly chaotic temporal patterns. Understanding the behavior of these time-ordered observations represents a critical step in decoding the underlying mechanisms governing numerous natural and artificial systems [2].

When considering real-world phenomena, time series data often contains multiple layers of information, including seasonal patterns, trending components, cyclic variations, and irregular fluctuations. These components collectively shape the observed temporal behavior and provide insights into the underlying processes driving the system dynamics [3]. Time series decomposition represents a fundamental analytical procedure that separates a time series into its constituent components, allowing for a deeper understanding of the underlying forces driving the observed data, making complex temporal patterns more interpretable and manageable for subsequent analysis [4].

The decomposition of time series data also plays a crucial role in distinguishing between deterministic and stochastic components, providing insights into the predictability and inherent variability of the observed phenomena [5]. By splitting systematic patterns from random fluctuations, decomposition techniques enable more accurate forecasting models that account for the different behaviors of each component. This separation is particularly valuable when developing models for phenomena with multiple time scales or mixed deterministic-stochastic characteristics, as it allows for the application of appropriate modeling techniques to each component based on its specific properties [6]. The hierarchical nature of many real-world processes can be effectively captured through decomposition, revealing how different temporal scales contribute to the overall observed behavior.

Time series decomposition techniques find applications across numerous domains. In economic forecasting, the Box-Jenkins ARIMA methodology utilizes time series decomposition principles to model and predict various economic indicators, such as foreign direct investment (FDI) inflows [7]. In natural resource management, time series decomposition provides valuable tools for analyzing and forecasting environmental parameters that influence resource availability and ecosystem health. For instance, research on water quality parameters has employed Box-Jenkins time series models alongside artificial neural networks to predict total dissolved solids in river systems, revealing that multilayer perceptron neural networks outperformed traditional time series approaches [8]. In demographic and public health domains, researchers have employed the Box-Jenkins approach to forecast male life expectancy using difference stationary processes and ARIMA models [9]. In the energy sector, time series decomposition plays a pivotal role in understanding and forecasting electricity demand patterns. The STL technique has been embedded within hidden Markov models with Gaussian processes to simulate high-resolution electricity demand data, providing insights into consumption patterns within community settings [3].

In structural health monitoring (SHM) for civil engineering structures, time series decomposition has become increasingly important. The wavelet transform, a powerful

form of time-frequency decomposition, has proven especially valuable in SHM for civil infrastructure [10]. By decomposing vibration signals and other structural response measurements into different frequency bands, engineers can identify specific patterns associated with damage or deterioration and distinguish them from normal operational variations or environmental influences. Wavelet-based techniques have been successfully applied to bridge assessments, facilitating the extraction of modal parameters and identification of subtle changes in dynamic behavior that may indicate emerging damage [11]. Advanced denoising algorithms based on discrete wavelet transform have demonstrated effectiveness in removing noise from vibration data, improving the accuracy of damage indicators, and reducing the influence of environmental and operational variability [12]. In cultural heritage buildings, SHM systems are routinely deployed to continuously monitor key structural parameters using networks of sensors that provide valuable insights into the dynamic behavior and long-term integrity of vulnerable masonry and stone structures, especially in seismically active areas [13, 14]. Methodologies using mixed static-dynamic long-term SHM systems incorporating crack meters, temperature sensors, and high-sensitivity accelerometers have enabled the detection of damage-induced anomalies in time series data, tracking both the amplitudes of major cracks and variations in natural frequencies to identify early signs of deterioration [15].

In order to exploit the hidden knowledge embedded in SHM data, this paper focuses on a specific case study involving the medieval wall of Gubbio, Italy, a region characterized by considerable seismic activity. Building upon the theoretical foundations discussed above, we propose a methodological framework to evaluate the capability of an inclinometer sensor, operating at a 20-minute sampling rate, to detect anomalies associated with seismic events. The methodology encompasses several sequential steps: first, a comprehensive analysis of the inclination time series to understand its inherent characteristics; second, the application of time series decomposition to separate the signal into its constituent components; and third, analysis of the residual component to identify potential anomalies that may correlate with seismic activities. To establish these correlations, we extract seismic event data through automated data scraping from the Italian National Institute of Geophysics and Volcanology (INGV) portal. The seismic data is then filtered for events relevant to the monitored structure's location, with appropriate transformations to estimate local ground motion parameters. By integrating inclinometer measurements with processed seismic data, this research aims to enhance our understanding of medieval masonry structures' responses to seismic excitations and evaluate the efficacy of inclinometer-based monitoring systems for heritage structures in seismically active regions.

2 Context and General Problem

Gubbio, located in central Italy, is a city where centuries of history are reflected in its urban landscape, shaped by both preservation efforts and ongoing change [16]. Its development began with the pre-Roman Umbrian civilization and was later transformed by Roman infrastructure such as the Via Flaminia, which redefined the city's layout

[16, 17]. The medieval period saw Gubbio flourish as a regional center, merging longstanding traditions with new urban strategies that continue to influence its character [16, 18].

The city’s medieval walls provide a vivid record of these historical processes. Some sections may have originated as early flood barriers, later extended in the 14th century to support defense and urban growth [16, 17]. Built from local stone and a variety of mortars, the walls reach up to 12 meters in height and vary in thickness from 0.5 to 3 meters [17, 18]. Over time, repeated repairs and modifications have produced a patchwork of materials and techniques, resulting in structural heterogeneity and discontinuities [18]. Environmental factors such as heavy rainfall, soil pressure, and temperature swings have accelerated deterioration, sometimes leading to partial collapses and instability [19]. Recent monitoring, using both satellite radar and on-site measurements, has detected seasonal movement and significant stress, emphasizing the need for ongoing conservation [18, 19].

Today, the condition of the walls is uneven. Some areas, like the stretch from Porta S. Ubaldo to Bughetto, have been stabilized through restoration, while others (particularly near Ranghiasi Park and the Cassero) show exposed cores, missing outer layers, and weakened joints. In these vulnerable spots, rainfall and hillside runoff increase loading and moisture, while plant growth accelerates both chemical and physical decay [18, 19].

To address these risks, a permanent monitoring system was installed in 2018 as part of the HERACLES project, which aims to strengthen heritage resilience to climate-related threats [20]. The system covers three key wall segments (Fig. 1) and uses monoaxial inclinometers (SEIKA NB2) to track out-of-plane motion, along with thermo-hygrometers (EE060) for temperature and humidity. Sensors record data every 10 minutes, transmitting wirelessly to a central gateway, and the readings are aggregated into 20-minute intervals for analysis. This dataset forms the basis for the present study [19].

For this study, analysis focused on data from monitoring station 02, a segment previously examined in related research and thus well-documented for comparative purposes. To ensure sufficient temporal coverage for detecting annual trends and seasonal effects, the dataset was limited to the first three complete years of monitoring, spanning from 2019 to 2021. The raw data, originally organized in daily files, were consolidated into a continuous time series. Inclination measurements were corrected for temperature effects according to the manufacturer’s guidelines, ensuring data reliability. After preprocessing and restricting the dataset to the selected period, the final cleaned dataset comprised 69,592 records, which served as the foundation for the subsequent time series decomposition and anomaly detection analyses.

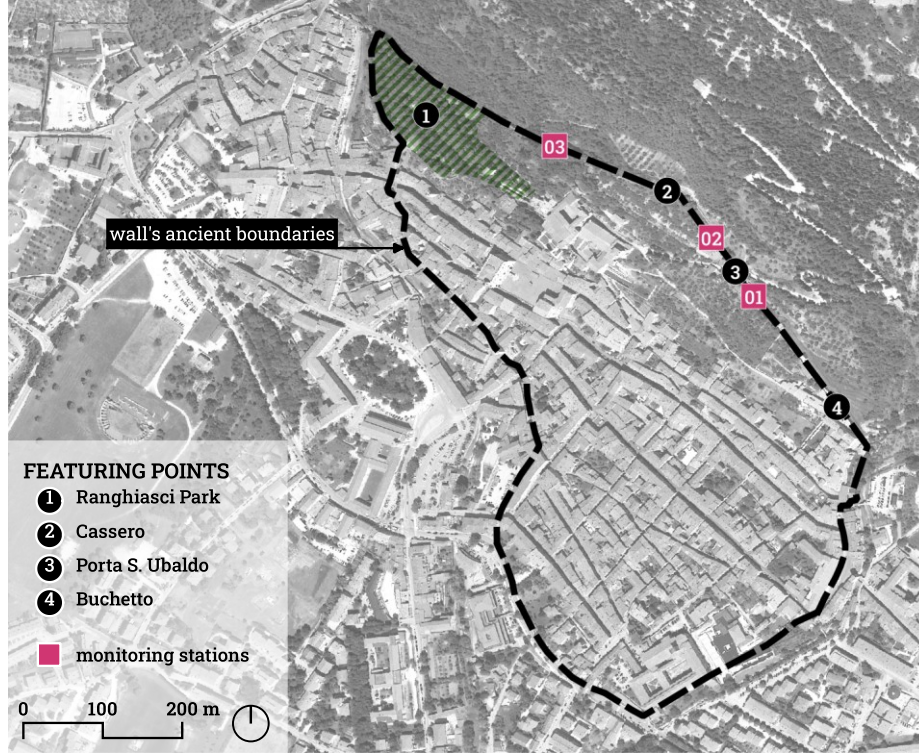


Fig. 1. Map of the wall's ancient boundaries and sections, and monitoring stations' location. Based on [18, 21].

3 Methodology

The methodology adopted in this study is structured in 3 main parts: (1) data exploration and preparation, which focuses on understanding the characteristics of the time series and addressing data quality issues, (2) time series decomposition, which aims to separate the signal into its constituent components for anomaly detection, and (3) creation of a dataset of significant seismic events, which aims to process a database of all the seismic events registered in the chosen time frame to individuate the ones could had pose a potential risk for the structure in question.

3.1 Data Exploration and Preparation

The preliminary stage of the analysis involved a comprehensive exploration of the inclinometer time series data. Initial visualization of the data through temporal scatterplots revealed noticeable discontinuities in the time series (Fig. 2), prompting a systematic investigation of data completeness. Given the expected sampling rate of 20 minutes over the three-year period, the dataset should theoretically contain 78,912 observations. However, only 69,592 records were present, indicating 9,320 missing

data points, approximately 11.8% of the expected total. This significant proportion of missing data necessitated a detailed gap analysis to inform the subsequent data preparation strategy.

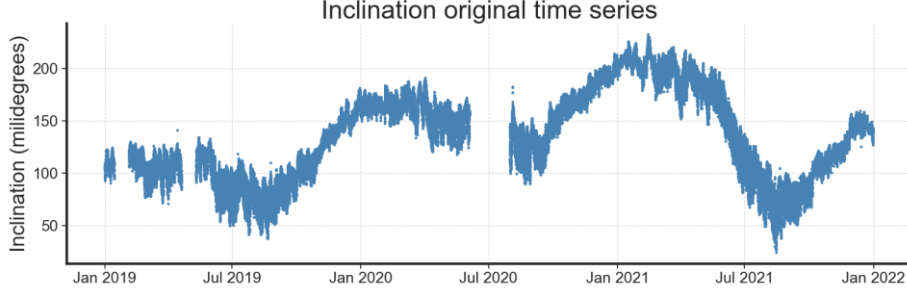


Fig. 2. Inclination original time series scatterplot.

To characterize the pattern of missing data, a comprehensive gap analysis was conducted, identifying each instance where consecutive observations were missing. The analysis revealed 877 distinct gaps distributed throughout the three-year monitoring period, with sizes ranging from single observations to extended periods of missing data. The statistical distribution of these gaps exhibited high positive skewness, with a mean gap size of 10.63 points and a substantially larger standard deviation of 156.09 points, indicating considerable variability in gap durations.

The frequency distribution of gap sizes showed a pronounced concentration at smaller intervals, with 72.1% of gaps consisting of single missing observations, 11.9% comprising two consecutive missing points, and 5.5% consisting of three points. This distribution pattern suggests that most data interruptions were brief, likely resulting from temporary sensor communication issues or data transmission failures. However, the dataset also contained several extensive gaps, with the three largest spanning 57 days (4,123 points), 20.7 days (1,493 points), and 20.5 days (1,475 points), respectively. These extended interruptions likely resulted from more significant system failures or maintenance operations.

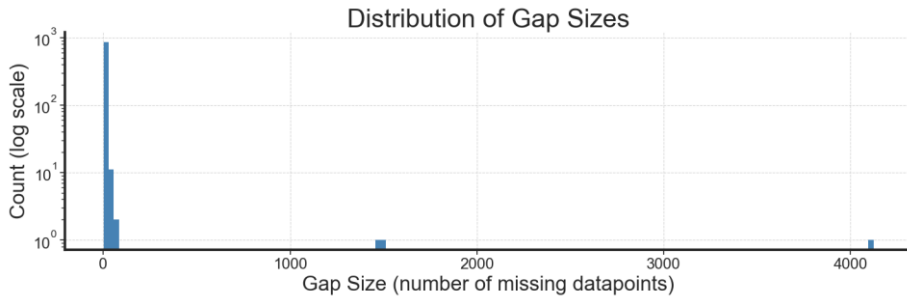


Fig. 3. Histogram of gap sizes

Given the heterogeneous nature of the gaps identified, a dual-approach imputation strategy was developed to address both the numerous small gaps and the few large interruptions. For small gaps (defined as ≤ 5 consecutive missing observations), a

simple linear interpolation method was employed. This approach assumes a linear trend between the values immediately before and after the gap, which is reasonable for short interruptions where the underlying physical process is unlikely to exhibit significant nonlinear behavior. This creates a prefilled dataset (Fig. 4), comprising only larger gaps, that could be processed with a second approach.

For larger gaps, particularly those exceeding 50 consecutive observations, a more sophisticated method was required to preserve the complex temporal characteristics of the inclination measurements. This study implemented the “Total imputation” method, as described by [22]. The approach extends the available data through backward, inverted replica before and after the original data sequence, creating a pattern that preserves the statistical properties of the original signal while filling gaps without introducing artificial discontinuities or distorting the underlying frequency characteristics.

From a technical perspective, the implementation of the Total imputation algorithm involves several critical procedures. When constructing the backward, inverted replicas, a crucial modification is applied: a linear slope adjustment is incorporated into each extension such that its terminal point aligns with the local mean at the beginning of the adjacent live data segment. This calibration is essential for preventing artificial discontinuities or “sawtooth” patterns at the boundaries where imputed data meets original measurements. The slope adjustment ensures smooth transitions while preserving the temporal characteristics of the original signal. The mathematical robustness of this approach derives from the fact that the extensions maintain independent, identically distributed (IID) properties relative to the original data. This preservation of statistical independence is achieved through three fundamental mechanisms: first, when calculating stability metrics such as Allan Deviation (ADEV), the processing typically occurs within T -length windows, while the Total extensions produce a periodicity of $2T$, which remains outside these analytical windows; second, the extended data are not artificially generated but rather consist of actual measured data that has been systematically transformed; and third, the IID characteristic persists in the extensions to the same degree as the measured noise is inherently IID, a property quantified as “self-similarity” in the time series.

Unlike traditional imputation approaches that deteriorate in performance with increasing gap size, the Total imputer can effectively address missingness patterns where total gaps exceed 100% of the data run. This robustness is especially relevant for our dataset, which contained several extended gaps, including a significant 57-day interruption. The method's validity has been empirically demonstrated through Monte Carlo trials, where it successfully recovered 30% missing data points with 90% confidence between original and recovered data across various power-law noise models. The results can be visualized in Fig. 4

Following the application of the dual imputation strategy, a comparative statistical analysis was conducted to assess the impact of the imputation on the distribution of the time series data. The comparison between the original and imputed datasets revealed several important characteristics. Despite the substantial addition of imputed values, the central tendency of the data remained remarkably stable (Table 1), with the mean value changing only by 0.85%, from 137.22 to 136.05. Similarly, the variability of the

data, as measured by the standard deviation, showed a modest decrease of 3.09%, from 41.33 to 40.05, suggesting a slight smoothing effect from the imputation process, an expected outcome as imputation methods typically cannot fully recreate the natural variability in missing segments.

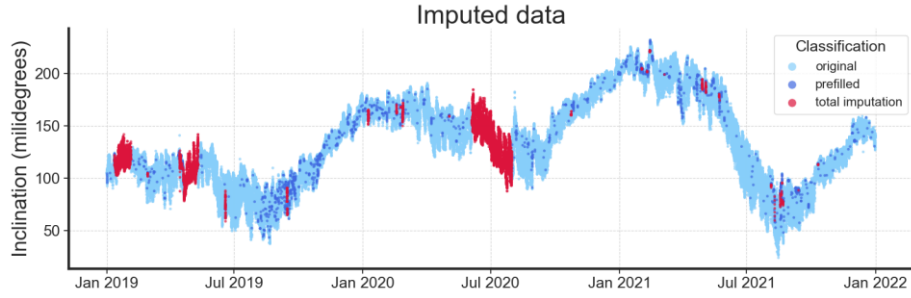


Fig. 4. Result of the hybrid imputation method.

Table 1. Statistical comparison between original and imputed dataset

Metric	Original data	Imputed data	% difference
Observations	69592	78912	+11.90
Mean	137.22	136.05	-0.85
Standard deviation	41.33	40.05	-3.09
Minimum value	23.72	23.72	0
25% Quartile	104.37	105.35	+0.93
50% Quartile	137.07	134.10	-2.17
75% Quartile	169.20	166.62	-1.52
Maximum value	232.52	232.52	0

The preservation of the distributional characteristics was further confirmed by comparing the quartiles of the original and imputed datasets. The first quartile (25th percentile) increased minimally from 104.38 to 105.35 (0.93%), the median decreased from 137.08 to 134.10 (-2.17%), and the third quartile (75th percentile) decreased from 169.20 to 166.63 (-1.52%). Importantly, the extreme values (minimum and maximum) remained identical in both datasets, indicating that the imputation did not introduce artificial extremes. This statistical stability suggests that the imputed values maintained the essential characteristics of the original data while successfully filling the identified gaps. The imputation quality assessment demonstrates that the combined linear interpolation and Total imputation approach successfully reconstructed the missing data without significantly altering the statistical properties of the original observations. With a complete and statistically sound dataset now established, we can proceed to develop a sophisticated decomposition framework capable of separating the complex temporal patterns within the inclinometer measurements, enabling the subsequent identification of potential seismic-induced anomalies in the residual component.

3.2 Time Series Decomposition

Following the comprehensive data exploration and gap filling, a sophisticated hybrid decomposition framework was developed and fine-tuned to the structural monitoring data from Gubbio's medieval wall. This framework addresses the complex thermal and seasonal dynamics observed in the inclination measurements, incorporating both additive and multiplicative relationships between components. The proposal investigated some data characteristics as a periodic signal, then proceeded to a hybrid decomposition to highlight the residuals, where some anomalies could be better seen.

The first step is to discover the hidden frequencies in the dataset. To do that, a magnitude spectrum of the gap-filled inclination series was obtained by transforming the discrete-time signal $y[n]$, $n = 0, \dots, N - 1$, into the frequency domain via its discrete Fourier transform (DFT) and then computing a one-sided amplitude spectrum. First, the DFT was computed using the fast Fourier transform (FFT) algorithm (Eq. 1) where N is the total number of samples and the sampling interval $\Delta t = 20$ min.

$$X[k] = \sum_{n=0}^{N-1} y[n] \exp(-2\pi i kn/N), k = 0, \dots, N - 1 \quad (1)$$

Next, the one-sided magnitude spectrum $M[k]$ was defined by normalizing the positive-frequency components of $X[k]$:

$$M[k] = \frac{|X[k]|}{N}, k = 1, \dots, [N/2] \quad (2)$$

To interpret spectral peaks in terms of periods T_k (hours), the frequency bins $f_k = k f_s/N$ were converted via

$$T_k = \frac{1}{f_k} = \frac{N \Delta t}{k}, k = 1, \dots, [N/2] \quad (3)$$

For visualization, $M[k]$ was plotted against T_k on a logarithmic axis, which highlights features across orders of magnitude in period. To guide interpretation, vertical markers were added at key periods: $T = 72$ observations (1 day), $T = 13,152$ observations (≈ 180 days), and $T = 26,304$ observations (≈ 365 days). This revealed distinct peaks at the key periods, quantitatively confirming the presence of diurnal thermal cycles and longer-term seasonal variations in the inclinometer data.

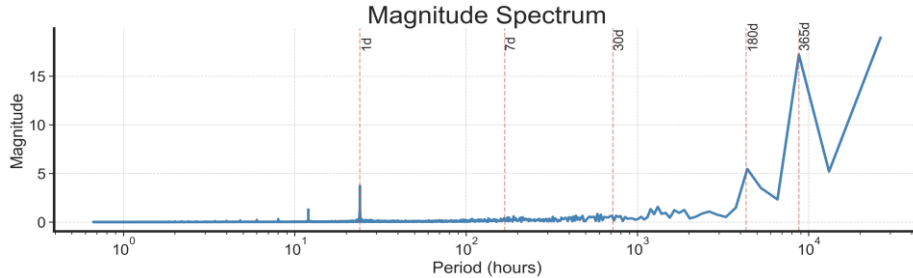


Fig. 5. Magnitude spectrum plot.

After that, an amplitude-variation analysis was carried out to characterize how the daily cycle itself changes over the year. For each day, the peak-to-peak amplitude was measured, then averaged across all years by both day-of-year and month. The resulting plot displays a marked winter–summer modulation: larger daily amplitudes in warmer months and smaller amplitudes during colder periods.

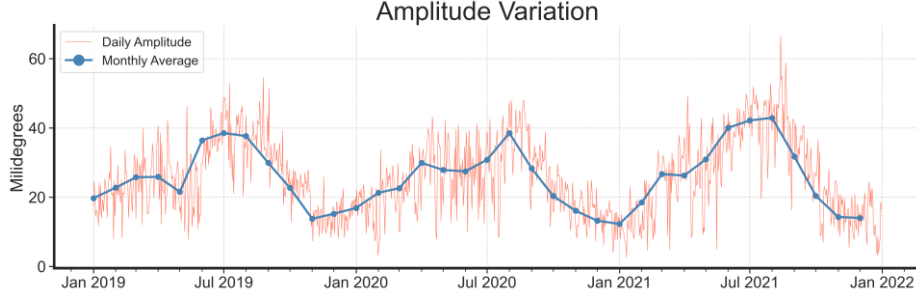


Fig. 6. Amplitude variation, averaged by day (red) and month (blue).

Observing that, it's possible to state that daily thermal cycles exhibit amplitude modulation according to seasonal patterns. This critical insight led to the development of a hybrid model where the daily component's amplitude is explicitly modulated by a smooth envelope function derived from observed seasonal variations. The mathematical formulation of this hybrid model can be expressed as:

$$y(t) = \text{Trend}(t) + \text{Annual}(t) + \text{Semi_annual}(t) + A(t) \times \text{Daily}(t) + \varepsilon(t) \quad (4)$$

where $y(t)$ represents the inclination time series, $\text{Trend}(t)$ captures the long-term drift, $\text{Annual}(t)$ and $\text{Semi_annual}(t)$ model the yearly and half-yearly cycles, $\text{Daily}(t)$ represents the diurnal thermal cycle, $A(t)$ is a smooth envelope function that modulates the daily cycle amplitude, and $\varepsilon(t)$ denotes the residual component containing unexplained variations and potential anomalies of interest.

The implementation of this decomposition proceeded through a two-stage adaptive ridge regression approach. In the first stage, we extracted the trend component using locally weighted scatterplot smoothing (LOWESS) [23] with a fractional parameter of 0.5, striking a balance between capturing meaningful long-term variations while excluding seasonal and daily fluctuations. This non-parametric approach is particularly suitable for structural monitoring data where the underlying drift may follow complex, non-linear patterns due to gradual settlement or long-term environmental influences.

The seasonal components were modeled using Fourier series representations with carefully selected periodicities. For the annual cycle, we employed the period of 26,304 observations, while the semi-annual component utilized the period of 13,152 observations. The daily component was modeled with a period of 72 observations, corresponding to our 20-minute sampling interval. The Fourier representation of these components can be generally expressed as:

$$C(t) = \sum_{k=1}^n \left[a_k \sin \left(\frac{2\pi kt}{P} \right) + b_k \cos \left(\frac{2\pi kt}{P} \right) \right] \quad (5)$$

where $C(t)$ represents a cyclical component, P is the period in number of observations, n is the Fourier order, and a_k and b_k are coefficients estimated later through ridge regression, in order to mitigate possible correlations between seasonalities.

For the daily component, rather than assuming a constant amplitude throughout the year, we quantified the daily amplitude capturing the 30-day centered moving average of the daily amplitude (Fig. 6), then modeling this smoothed pattern using a Fourier series representation (order=6) to generate a continuous envelope function $A(t)$. This envelope function was subsequently normalized by its mean to maintain scale interpretability.

The second stage of our decomposition employed adaptive regularization based on component-specific variability. Ridge regression, a form of L2-regularized linear regression [24], penalizes the magnitude of coefficient values to prevent overfitting. However, rather than applying uniform regularization across all components, we implemented an adaptive approach that adjusts the penalty strength inversely proportional to each component's temporal variability. This can be formalized as:

$$\alpha_j = \frac{\alpha_0}{\max(\text{Var}(\Delta C_j), \epsilon)} \quad (6)$$

where α_j is the regularization parameter for component j , α_0 is a base regularization strength, $\text{Var}(\Delta C_j)$ represents the variance of first differences of component j from the initial fit, and ϵ is a small constant to prevent division by zero. This adaptive regularization ensures that components with higher natural variability are not overly smoothed, while still providing sufficient regularization for more stable components.

The multiplicative interaction between the daily component and its envelope function was implemented by modulating the Fourier feature matrix. Each column of the daily Fourier feature matrix was multiplied by the corresponding envelope function value, creating a time-varying basis that captures the seasonal amplitude modulation. This transformed feature matrix was then used in the ridge regression, effectively implementing the multiplicative term $A(t) \times \text{Daily}(t)$ in our hybrid model.

To enhance robustness against potential outliers in the data, we employed an iterative reweighting scheme. After fitting an initial model with equal weights, we computed residuals and transformed them into robust z-scores using the median absolute deviation (MAD) as a scale estimator:

$$z_i = \frac{0.6745 \times (r_i - \text{median}(r))}{\text{MAD}} \quad (7)$$

where r_i represents the residual for observation i . Observations with absolute z-scores exceeding 3.5 were downweighted inversely proportional to their deviation, specifically using weights $w_i = 3.5/|z_i|$ for outliers. This approach effectively reduces

the influence of anomalous observations while preserving their information in the dataset.

The final model coefficients were determined through a weighted second ridge regression that incorporated both the adaptive regularization parameters and the outlier-robust weights. Additionally, since the framework was implemented in Python language, some manual component weight adjustment were inserted, providing flexibility to emphasize or de-emphasize specific components based on domain knowledge or analytical objectives. The final decomposition yielded separate time series for trend, annual cycles, semi-annual cycles, modulated daily cycles, and residuals, together with the envelope function quantifying the seasonal variation in daily amplitude.

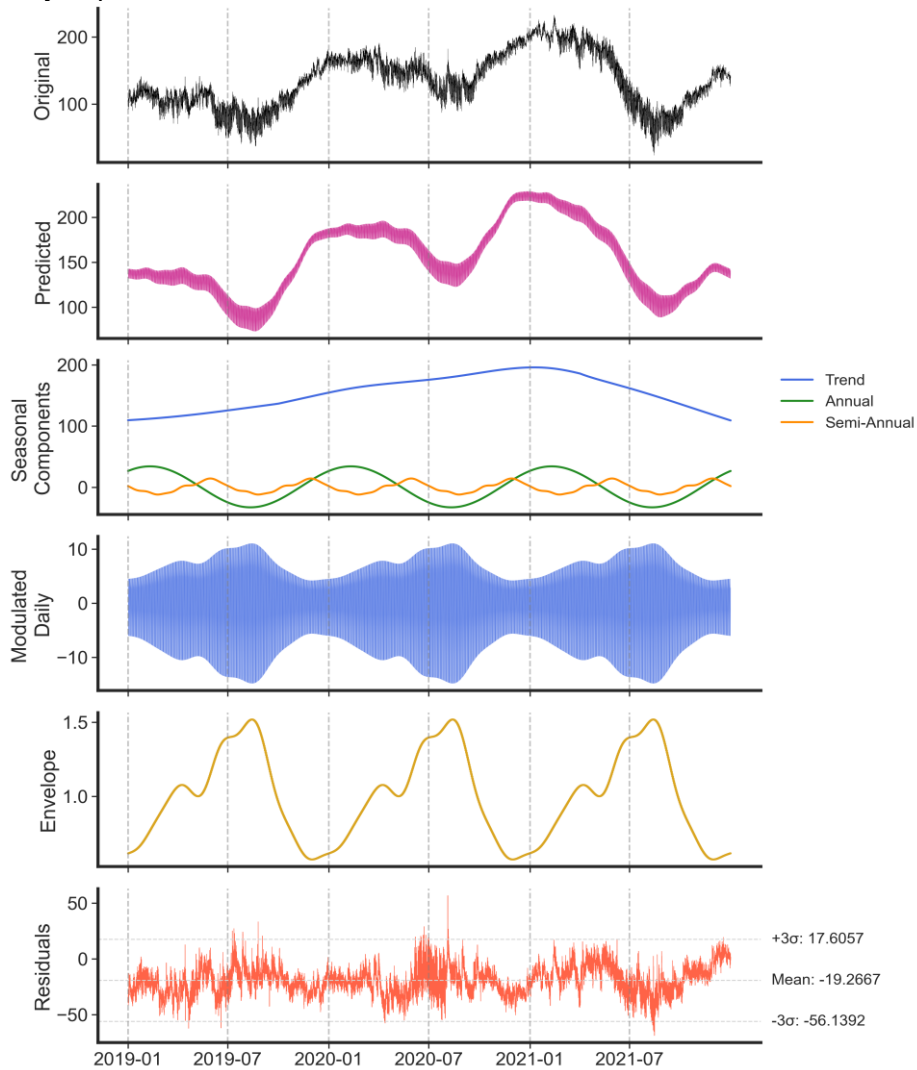


Fig. 7. Time Series decomposition.

Having successfully decomposed the inclination time series into its constituent components, the attention is now reserved to the third methodological component: creating and analyzing a database of significant seismic events that could potentially correlate with anomalies identified in the residual component of the decomposition.

3.3 Creation of a Dataset of Significant Seismic Events

The third methodological component focused on creating a comprehensive database of seismic events that could potentially correlate with anomalies in the residual component. This process involved automated data acquisition, geographic filtering, and engineering analysis to transform raw seismic catalog data into structural-specific impact metrics.

The acquisition of seismic data utilized an automated web scraping framework developed in Python to extract event records from the Italian National Institute of Geophysics and Volcanology (INGV) webservice portal. The data retrieval process employed time-windowed queries to manage server load, processing 30-day intervals sequentially. The geographical bounds were set to encompass all of Italy (latitude: 35.0 °-48.0 °N, longitude: 6.0 °-19.0 °E), ensuring comprehensive coverage of the seismic activity that could potentially affect the Gubbio medieval walls. For each event, critical parameters were collected, including timestamp, geographic coordinates (latitude, longitude), depth, magnitude, and event ID.

A filter was applied to isolate events with potential relevance to Gubbio's medieval walls. This process utilized a comprehensive geospatial analysis framework converting standard earthquake data into a GeoDataFrame for advanced spatial operations. The epicentral distance between each seismic event and the monitored structure (coordinates: 43.35 °N, 12.58 °E) was calculated using the Haversine formula [25] shown in Eq. 8:

$$d_{\text{epi}} = 2r \cdot \arcsin \left(\sqrt{\sin^2 \left(\frac{\phi_2 - \phi_1}{2} \right) + \cos(\phi_1) \cos(\phi_2) \sin^2 \left(\frac{\lambda_2 - \lambda_1}{2} \right)} \right) \quad (8)$$

where d_{epi} represents the epicentral distance, r is Earth's radius, and ϕ and λ are the latitudes and longitudes of the two points (earthquake epicenter and structure), respectively. The hypocentral distance was subsequently calculated using the Pythagorean relation:

$$d_{\text{hyp}} = \sqrt{d_{\text{epi}}^2 + d_{\text{depth}}^2} \quad (9)$$

where d_{hyp} is the hypocentral distance, d_{epi} is the epicentral distance, and d_{depth} is the earthquake focal depth.

For each event, ground motion parameters at the wall location were estimated using Kawasumi formula [26]. The primary parameter of interest was Peak Ground Acceleration (PGA), calculated as:

$$\log(a) = M_s - 5.4 - 0.00084(d_{\text{hyp}} - 100) + \frac{\log_{10}(100/d_{\text{hyp}})}{0.43429} \quad (10)$$

where a is the PGA in cm/s^2 , M_s is magnitude, and d_{hyp} is hypocentral distance in kilometers.

To account for nonlinear behavior at high magnitudes, a magnitude-dependent attenuation factor for events where $M > 6.0$ was applied using the empirically observed saturation term a [27]:

$$\log(a) = -1.72 + 0.43M \quad (4)$$

The seismic events were classified into significance categories based on empirical relationships between Peak Ground Acceleration (PGA) and Modified Mercalli Intensity (MMI), as established in the literature through regression analyses and tabulated data, linking specific PGA ranges to observed intensity and typical effects [28, 29]. Table 2 summarizes the adopted classification.

Table 2. PGA-MMI Relationships Across Multiple Studies

Category	PGA (m/s^2)	According to [28]	According to [29]	Typical Effects
Negligible	< 0.1	Below detection threshold	Below detection threshold	Felt by few, no damage
Low	$0.1 - 0.5$	$0.1-0.5 \text{ m/s}^2$ (MMI V)	0.3 m/s^2 (MMI V)	Felt by most, minor non-structural damage
Medium	$0.5 - 2.5$	$0.5-2.5 \text{ m/s}^2$ (MMI VI-VII)	0.6 m/s^2 (MMI VI), 1.2 m/s^2 (MMI VII)	Damage to weak structures
High	≥ 2.5	$>2.5 \text{ m/s}^2$ (MMI VIII-IX)	2.3 m/s^2 (MMI VIII), 4.5 m/s^2 (MMI IX)	Structural damage, collapse possible

This dataset was then temporally aligned with the residual component from the time series decomposition for subsequent correlation analysis between seismic events and trend changes.

4 Results

The residual component plot reveals substantial variations in the structure's inclination after removing trend, seasonal, and temperature effects (Fig. 8). The residuals exhibit a negative mean (-19.28 millidegrees), indicating a persistent bias in one direction. Several periods demonstrate notable deviations beyond the statistical control limits ($\pm 3\sigma$), particularly:

- July 2020: An exceptional positive excursion reaching nearly 60 millidegrees, coinciding with a period marked by a low-intensity seismic event

- September 2021: A significant negative deviation approaching -70 millidegrees, occurring shortly after a high-intensity seismic event
- Multiple intermediate excursions throughout the monitoring period that repeatedly breach the $\pm 3\sigma$ thresholds

The overall pattern suggests that while the decomposition model effectively captured most systematic variations, the residuals retain meaningful signal components potentially associated with structural responses to external forces.

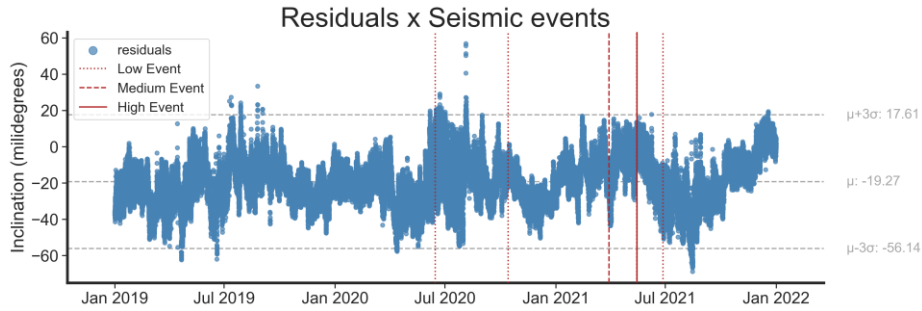


Fig. 8. Raw inclination residuals overlaid by relevant seismic events.

The changes in residuals (first-order differences) provide insight into the immediate dynamic response of the structure (Fig. 9). Most changes remain within the established statistical bounds (± 6.39 millidegrees), suggesting that under normal conditions, the structure exhibits relatively stable behavior with consistent variance. However:

- Several outlier points exceed these thresholds throughout the monitoring period
- A particularly significant positive change appears in late 2021, representing an abrupt shift in structural behavior
- The pattern of changes shows no immediate visual one-to-one correlation with marked seismic events, suggesting a complex, possibly delayed response mechanism

This differential analysis reveals that the structure typically responds gradually rather than exhibiting sudden large displacements, even following seismic activity.

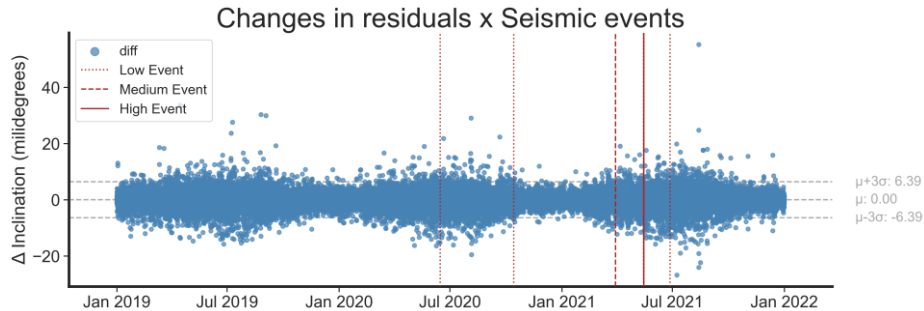


Fig. 9. First-order differences of the inclination residuals overlaid by relevant seismic events.

The Cumulative Sum analysis (Fig. 10) provides the most compelling evidence of potential seismic influence on structural behavior. This technique, which tracks accumulated deviations over time, reveals:

- Multiple distinct behavioral regimes throughout the monitoring period, indicated by alternating periods of increasing (CUSUM+) and decreasing (CUSUM-) shifts
- A substantial increasing shift in mid-2021 that develops shortly after a high-intensity seismic event, suggesting a possible cause-effect relationship
- An even more dramatic positive shift near the end of 2021, approaching a CUSUM statistic of 2000, potentially indicating a significant change in the structure's condition
- A notable decreasing shift in early-to-mid 2020 approaching a CUSUM statistic of -500, which may represent another important behavioral transition

The CUSUM chart effectively amplifies subtle persistent changes that might be overlooked in the raw residuals, highlighting potential intervention points in the structure's behavior that merit further investigation.

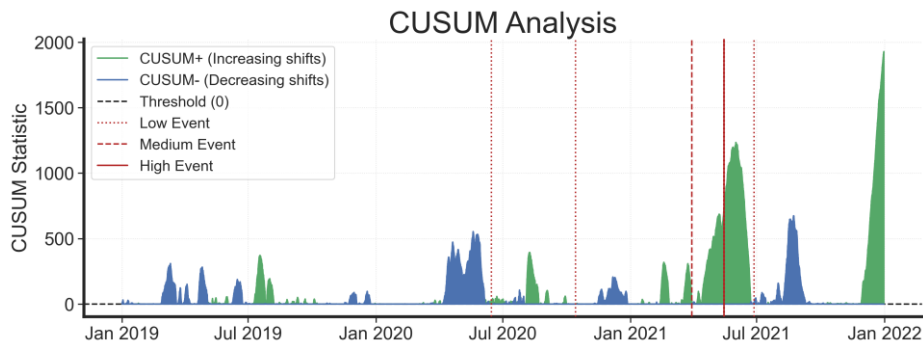


Fig. 10. CUMSUM analysis overlaid by relevant seismic events.

When analyzed collectively, these visualizations suggest that while immediate large-scale responses to seismic events are not consistently evident, there appear to be delayed or cumulative effects that manifest as persistent shifts in structural behavior. The high-intensity seismic event in mid-2021 precedes a period of increased variability and ultimately a large positive CUSUM shift, suggesting that seismic activity may trigger progressive changes in structural response rather than immediate dramatic shifts.

The analysis also reveals that the structure exhibits complex dynamic behavior with multiple timescales of response—immediate changes captured in the first-order differences, medium-term fluctuations visible in the raw residuals, and long-term persistent shifts highlighted by the CUSUM analysis. This multi-timescale response mechanism provides valuable insight into how medieval masonry structures might respond to and accumulate the effects of seismic excitations over time.

5 Conclusion

This research has shown the potential of advanced time series decomposition techniques for structural health monitoring of heritage masonry structures, specifically applied to the medieval city walls of Gubbio, Italy. Through a hybrid decomposition approach, an inclinometer signal was split into trend, seasonal, envelope, and residual components, enabling the isolation of potential structural responses to external stimuli. The analysis of residuals revealed significant excursions beyond statistical control limits, with a persistent negative bias (-19.28 millidegrees) suggesting an underlying directional tendency in the structure's behavior. First-order differencing further illuminated the dynamic characteristics of the wall's response mechanism, indicating that the structure typically exhibits gradual rather than abrupt displacements, even following seismic events. Most notably, the CUSUM analysis provided compelling evidence of regime shifts in structural behavior, with substantial cumulative deviations occurring in temporal proximity to seismic events, particularly the high-intensity event of mid-2021, which preceded a dramatic positive shift in the CUSUM statistic approaching 2000. These findings suggest that medieval masonry structures may manifest delayed or progressive responses to seismic excitations, accumulating effects over multiple timescales rather than displaying immediate dramatic shifts.

The methodology developed holds significant potential for enhancing heritage conservation practices by enabling early detection of subtle changes in structural behavior before visible damage occurs. Furthermore, the multi-timescale analysis framework provides valuable insights into the complex dynamic behavior of historical masonry structures under environmental and seismic influences.

Future research should focus on applying alternative decomposition methodologies to exploit potential better capture of non-stationary characteristics in the time series data. The exploration of environmental data could identify additional excitation sources affecting structural response. Multivariate modeling incorporating these parameters might explain anomalies currently not explained by seismic activity, potentially revealing complex environmental-structural interactions. Integrating finite element modeling would validate hypothesized response mechanisms and allow simulation of various loading scenarios, helping distinguish between localized and global structural behaviors while quantifying stress distributions. This combined approach of advanced time-frequency analysis, comprehensive environmental assessment, and physics-based simulation would create a more robust framework for interpreting subtle anomalies in heritage structure monitoring data, enhancing preservation decision-making for historically significant masonry structures.

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